



Working Draft

MEF W122.1v0.2

LSO Cantata and LSO Sonata Site Management API - Developer Guide

This draft represents MEF work in progress and is subject to change.

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List of Contributing Members

The following members of the MEF participated in the development of this document and have requested to be included in this list.

Member

Table 1. Contributing Members

1. Abstract

This standard is intended to assist in the implementation of the Site Retrieval functionality defined for the LSO Cantata and LSO Sonata Interface Reference Points (IRPs), for which requirements and use cases are defined in *Installation Place and Service Site Management Business Requirements and Use Cases* [MEF150].

Site retrieval allows the Buyer to determine if there is already a Site, as defined by the Seller, at the Geographic Address. The Buyer requests a list of Sites at a Geographic Address or, if the Site **id** is known, may ask for details of that Site. Site information includes the type of Site (Public or Private), the specific location within the Geographic Address where the Site is located, and additional Site information. The Buyer can use the Site **id** for POQ, Quote, or Product Order.

This standard normatively incorporates the following files by reference as if they were part of this document, from the GitHub repository

<https://github.com/MEF-GIT/MEF-LSO-Sonata-SDK>

- `productApi/serviceability/site/geographicSiteManagement.api.yaml`

<https://github.com/MEF-GIT/MEF-LSO-Cantata-SDK>

- `productApi/serviceability/site/geographicSiteManagement.api.yaml`

2. Terminology and Abbreviations

This section defines the terms used in this document. In many cases, the normative definitions of terms are found in other documents. In these cases, the third column is used to provide the reference that is controlling, in other MEF or external documents.

In addition, terms defined in the standards referenced below are included in this document by reference and are not repeated in the table below:

- [MEF 50.1](#)
- [MEF 55.1](#)
- [MEF 55.1.1](#)
- [MEF 150](#)

Term	Description	Reference
Fielded Address Representation	A type of Representation where a discrete field and value for each type of boundary or identifier down to the lowest level of detail. For example, "street number" is one field, "street name" is another field, etc.	This document; adapted from [MEF 150]
Formatted Address Representation	A type of Representation using single string based on local postal addressing conventions.	This document; adapted from [MEF 150]
Geographic Address	A place on Earth, which may or may not be fixed, described using one or more Geographic Address Representations.	This document; adapted from [MEF 150]
Geographic Address Representation	A concrete method of describing a specific address which uses well defined formats to detail the attributes.	This document; adapted from [MEF 150]
Geographic Point Representation	A type of Representation where coordinates (latitude, longitude and sometimes elevation) are used to specify a particular place on Earth.	This document; adapted from [MEF 150]
Geographic Site	A fixed or mobile place at which a Product can be installed.	This document; adapted from [MEF 150]

Label Representation	A type of Geographic Address Representation that is a unique combination of label and Administrative Authority (any organization that distributes labels) that controls assignment of the label and that specifies either a place which may or may not be fixed on Earth.	This document; adapted from [MEF 150]
OFF-NET	A Site that is served by a partner of the Seller and not directly by the Seller's network.	This document; adapted from [MEF 150]
ON-NET	A Site that is connected to the Seller's network.	This document; adapted from [MEF 150]
Private Site	A Geographic Site for which the existence is on a need-to-know basis.	This document; adapted from [MEF 150]
Public Site	A Geographic Site for which the existence is public information.	This document; adapted from [MEF 150]
Requesting Entity	The business organization that is acting on behalf of one or more Buyers. In the most common case, the Requesting Entity represents only one Buyer and these terms are then synonymous.	[MEF 150]
Responding Entity	The business organization that is acting on behalf of one or more Sellers. In the most common case, the Responding Entity represents only one Seller and these terms are then synonymous.	[MEF 150]
REST API	Representational State Transfer. REST provides a set of architectural constraints that, when applied as a whole, emphasizes scalability of component interactions, generality of interfaces, independent deployment of components, and intermediary components to reduce interaction latency, enforce security, and encapsulate legacy systems.	[REST]

Table 2. Terminology

3. Compliance Levels

The key words "**MUST**", "**MUST NOT**", "**REQUIRED**", "**SHALL**", "**SHALL NOT**", "**SHOULD**", "**SHOULD NOT**", "**RECOMMENDED**", "**NOT RECOMMENDED**", "**MAY**", and "**OPTIONAL**" in this document are to be interpreted as described in BCP 14 (RFC 2119 [[rfc2119](#)], RFC 8174 [[rfc8174](#)]) when, and only when, they appear in all capitals, as shown here. All key words must be in bold text.

Items that are **REQUIRED** (contain the words **MUST** or **MUST NOT**) are labeled as **[Rx]** for required. Items that are **RECOMMENDED** (contain the words **SHOULD** or **SHOULD NOT**) are labeled as **[Dx]** for desirable. Items that are **OPTIONAL** (contain the words **MAY** or **OPTIONAL**) are labeled as **[Ox]** for optional.

A paragraph preceded by **[CRa]<** specifies a conditional mandatory requirement that **MUST** be followed if the condition(s) following the "<" have been met. For example, "**[CR1]<[D38]**" indicates that Conditional Mandatory Requirement 1 must be followed if Desirable Requirement 38 has been met. A paragraph preceded by **[CDb]<** specifies a Conditional Desirable Requirement that **SHOULD** be followed if the condition(s) following the "<" have been met. A paragraph preceded by ****[COc]<*** specifies a Conditional Optional Requirement that **MAY** be followed if the condition(s) following the "<" have been met.

4. Introduction

Note: The requirements and use cases for Site retrieval functionality are defined in *Installation Place and Service Site Management Business Requirements and Use Cases* [MEF150] which introduced the *Installation Place* as a new entity with multiple address representations. For the sake of backward and TMF compatibility the API definition does not introduce *Installation Place* as a new API resource, but updates the existing model of *GeographicAddress*, according to the before-mentioned MEF Standard.

This standard is based on TMF 674 API as specified by *TMF674 Geographic Site Management API User Guide* [TMF674].

This standard specification document describes the API for Site Retrieval functionality of the LSO Cantata Interface Reference Point (IRP) and Sonata IRP as defined in the MEF 55.1 *Lifecycle Service Orchestration (LSO): Reference Architecture and Framework* [MEF55.1]. The LSO Reference Architecture is shown in Figure 1 with both IRPs highlighted.

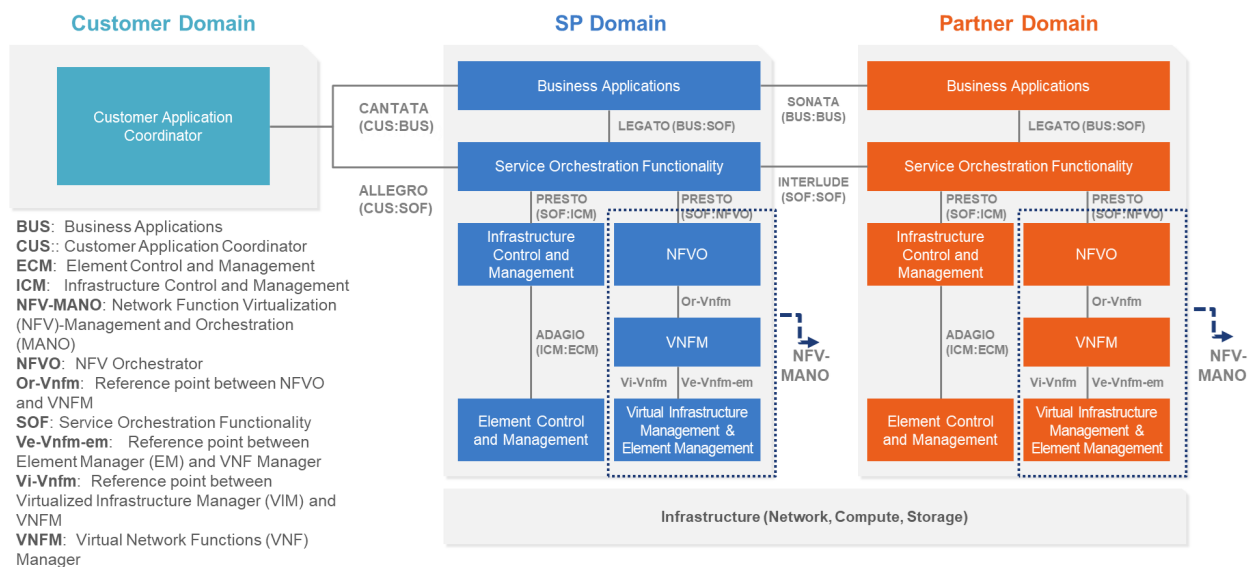


Figure 1. The LSO Reference Architecture

Cantata and Sonata IRPs define pre-ordering and ordering functionalities that allow an automated exchange of information between business applications of the Buyer (Customer or Service Provider) and Seller (Service Provider or Partner) Domains. Those are:

- Product Catalog
- Address Validation
- Site Retrieval
- Product Offering Qualification
- Quote
- Product Offering Availability and Pricing Discovery
- Product Inventory
- Product Ordering
- Trouble Ticketing
- Billing

The business requirements and use cases for Site Retrieval are defined in *Installation Place and Service Site Management Business Requirements and Use Cases* [MEF150].

This document focuses on the implementation aspects of Site Retrieval functionality and is structured as follows:

- [Chapter 4](#) provides an introduction to Site Retrieval and its description in a broader context of Cantata and Sonata and their corresponding SDKs.
- [Chapter 5](#) gives an overview of endpoints, resource model and design patterns.
- Use cases and flows are presented in [Chapter 6](#).
- And finally, [Chapter 7](#) complements previous sections with a detailed API description.

4.1. Description

A Site usually represents a location where the Seller has already delivered one or more products. A Site identifier is assigned at some point by the Seller to reference the location. A Site always references a [GeographicAddress](#) that defines its location.

A [GeographicSite](#) within a Seller's network represents a location at which a Product can be installed and usually where the Seller has already delivered one or more products. A [GeographicSite.id](#) is assigned by the Seller to reference the [GeographicSite](#). A [GeographicSite](#) is associated to one and only one [GeographicAddress](#).

Before performing Product Offering Qualification, Quote, or Product Order, the Buyer may need to obtain details on the Site. These details can include whether the Site is public or private, room and floor information, or additional details.

One example of why a Buyer may need to perform a Site retrieval prior to submitting a Product Offering Qualification request is to determine if that site is public, meaning that the Seller is willing to provide service to any end-customer at the site, or private, meaning that the end-customers that the Seller will provide service to at that site is limited.

Public Sites are often located in a common area at a Geographic Address, such as a basement, which all end-customers at that Geographic Address can access. Normally, all public Sites at a Geographic Address will be included in the list of Sites returned by the Seller in response to the Buyer's retrieve Site list request.

Private Sites are dedicated to a single end customer and are not usually shared with other end customers at that Geographic Address. The Seller cannot connect a Buyer to the private Site without the permission of a party authorized to use the private Site. Normally, a private Site will not be included in the list of Sites at a Geographic Address returned by the Seller in response to the Buyer's retrieve Site list request unless the Buyer is determined by the Seller as authorized to view the private Sites at that Geographic Address.

A Site is defined as a physical place at which a Product can be installed. There are many possible cases for example: A particular Geographic Address may not have any suitable Sites to install telecom equipment; it may have one Site (e.g., a wiring closet in the basement); or it may have multiple Sites (e.g., a wiring closet in each suite of a multi-tenant building). Moreover, even if the Seller is aware of the Geographic Address, they may or may not hold any of this Site information at the time of the Buyer's request.

4.2. Conventions in the Document

- Code samples are formatted using code blocks. When notation `<< some text >>` is used in the payload sample it indicates that a comment is provided instead of an example value and it might not comply with the OpenAPI definition.
- Model definitions are formatted as in-line code (e.g. [GeographicSite](#)).
- In UML diagrams the default cardinality of associations is `0..1`. Other cardinality markers are compliant with the UML standard.

- In the API details tables and UML diagrams required attributes are marked with a * next to their names.
- In UML sequence diagrams `{{variable}}` notation is used to indicate a variable to be substituted with a correct value.

4.3. Approach

As presented in Figure 2. both Cantata and Sonata API frameworks consists of three structural components:

- Generic API framework
- Product-independent information (Function-specific information and Function-specific operations)
- Product-specific information (MEF product specification data model)

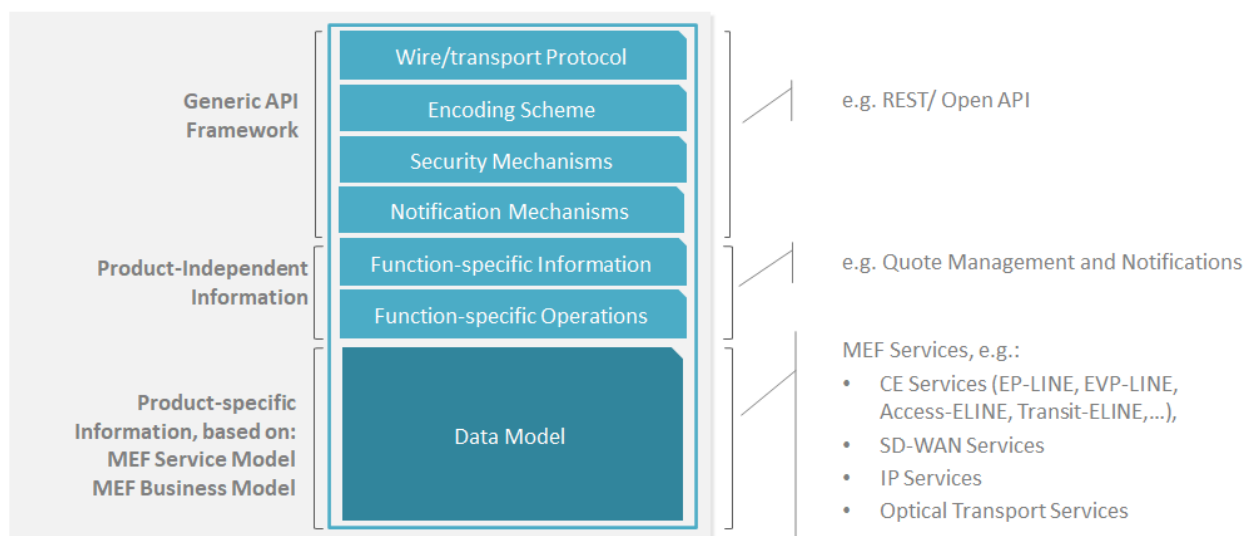


Figure 2. Cantata and Sonata API framework

The essential concept behind the framework is to decouple the common structure, information and operations from the specific product information content.

Firstly, the Generic API Framework defines a set of design rules and patterns that are applied across all Cantata or Sonata APIs.

Secondly, the product-independent information of the framework focuses on a model of a particular Cantata or Sonata functionality and is agnostic to any of the product specifications.

Finally, the product-specific information part of the framework focuses on MEF product specifications that define business-relevant attributes and requirements for trading MEF subscriber and MEF operator services.

The Site Retrieval is product-agnostic in its nature and is not intended to carry any product-specific information. It operates using the Generic API Framework and the Function-Specific Information and Operations.

4.4. High-Level Flow

Site Retrieval is part of a broader Cantata and Sonata End-to-End flow. Figure 3. below shows a high-level diagram to get a good understanding of the whole process and Site Retrieval's position within it.

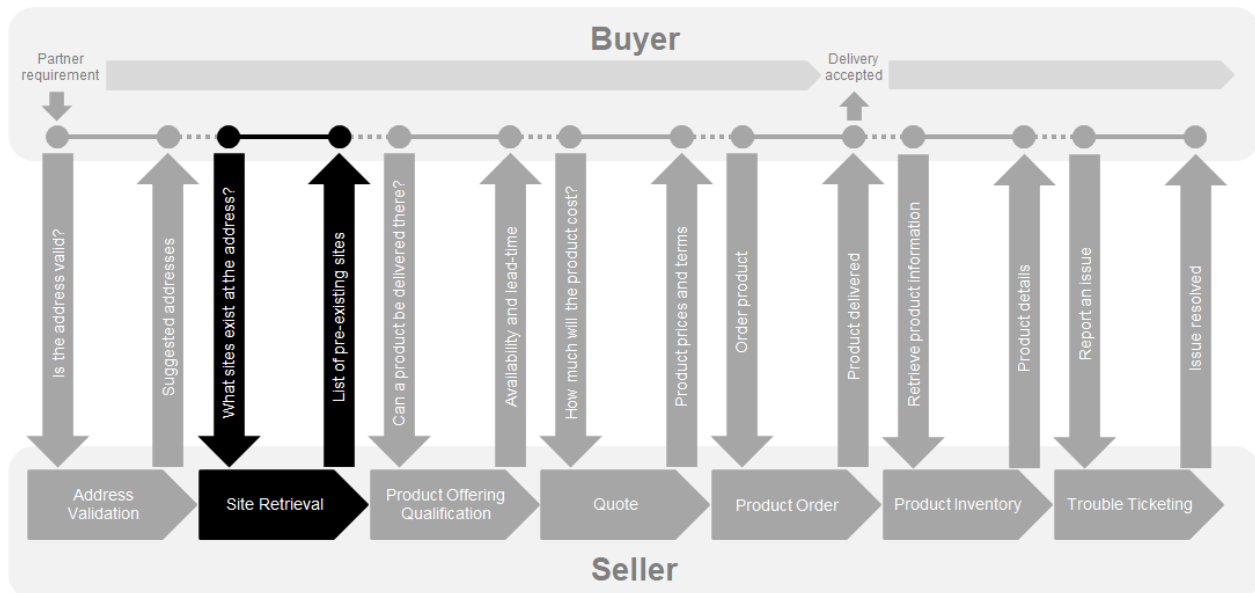


Figure 3. Cantata and Sonata End-to-End Function Flow

- **Address Validation:**
 - Allows the Buyer to retrieve address information from the Seller, including exact formats, for Geographic Addresses known to the Seller.
- **Site Retrieval:**
 - Allows the Buyer to retrieve Geographic Site information including exact formats for Geographic Sites known to the Seller.
- **Product Offering Qualification (POQ):**
 - Allows the Buyer to check whether the Seller can deliver a product or set of products from among their product offerings at the geographic address or a Geographic Site specified by the Buyer; or modify a previously purchased product.
- **Quote:**
 - Allows the Buyer to submit a request to find out how much the installation of an instance of a Product Offering, an update to an existing Product, or a disconnect of an existing Product will cost.
- **Product Order:**
 - Allows the Buyer to request the Seller to initiate and complete the fulfillment process of an installation of a Product Offering, an update to an existing Product, or a disconnect of an existing Product at the address defined by the Buyer.
- **Product Inventory:**
 - Allows the Buyer to retrieve the information about existing Product instances from Seller's Product Inventory.
- **Trouble Ticketing:**
 - Allows the Buyer to create, retrieve, and update Trouble Tickets as well as receive notifications about Incidents' and Trouble Tickets' updates. This allows for managing issues and situations that are not part of the normal operations of the Product provided by the Seller.

5. API Description

This section discusses the API structure and design patterns. It starts with the high-level use cases diagram and then it describes the REST endpoints with the use case mapping.

5.1. High-level use cases

Figure 4 presents a high-level use case diagram. This picture aims to help understand endpoint mapping. Use cases are described extensively in [chapter 6](#). Use case numbering is kept consistent with [MEF 150](#). The underlined font designates required use cases.

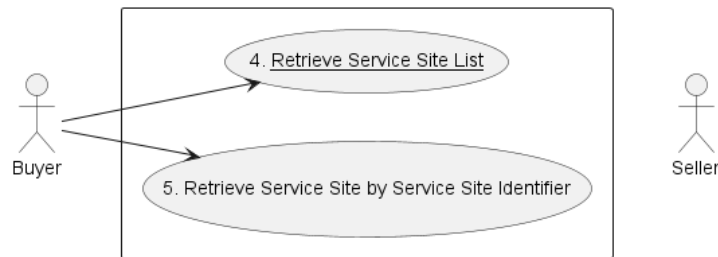


Figure 4. High-level use cases

5.2. Resource/endpoint Description

5.2.1. Seller Side Endpoints

Base URL for Cantata:

```
https://{{serverBase}}:{{port}}
{{?/seller_prefix}}/mefApi/cantata/geographicSiteManagement/v2/
```

Base URL for Sonata:

```
https://{{serverBase}}:{{port}}
{{?/seller_prefix}}/mefApi/sonata/geographicSiteManagement/v8/
```

Note: All examples will include only the Sonata version of the Base Path.

The following endpoints are exposed by the Seller and allow the Buyer to:

- perform a query for a Geographic Site list
- get a single Geographic Site by **id**

The endpoints and corresponding data model are defined in [productApi/serviceability/site/geographicSiteManagement.api.yaml](#).

API endpoint	Description	MEF 150 Use case Mapping
GET /geographicSite	A request initiated by the Buyer to retrieve a list of GeographicSites from the Seller based on filter criteria provided as query	UC 4: Retrieve Service Site List

API endpoint	Description	MEF 150 Use case Mapping
GET /geographicSite/{id}	A request initiated by the Buyer to retrieve full details of a single GeographicSite based on a Geographic Site identifier previously provided by the Seller.	UC 5: Retrieve Service Site by Identifier

Table 3. Endpoint to use case mapping

[R1] The Buyer **MUST** be able to use the GET /geographicSite operation to perform Use Case 4. [MEF150 R21]

5.2.2. Specifying the Buyer ID and the Seller ID

A business Entity willing to represent multiple Buyers or multiple Sellers must follow requirements of [MEF 150] chapter 8.8, which states:

For requests of all types, there is a business Entity that is initiating an Operation (called a Requesting Entity) and a business Entity that is responding to this request (called the Responding Entity). In the simplest case, the Requesting Entity is the Buyer and the Responding Entity is the Seller. However, in some cases, the Requesting Entity may represent more than one Buyer and similarly, the Responding Entity may represent more than one Seller.

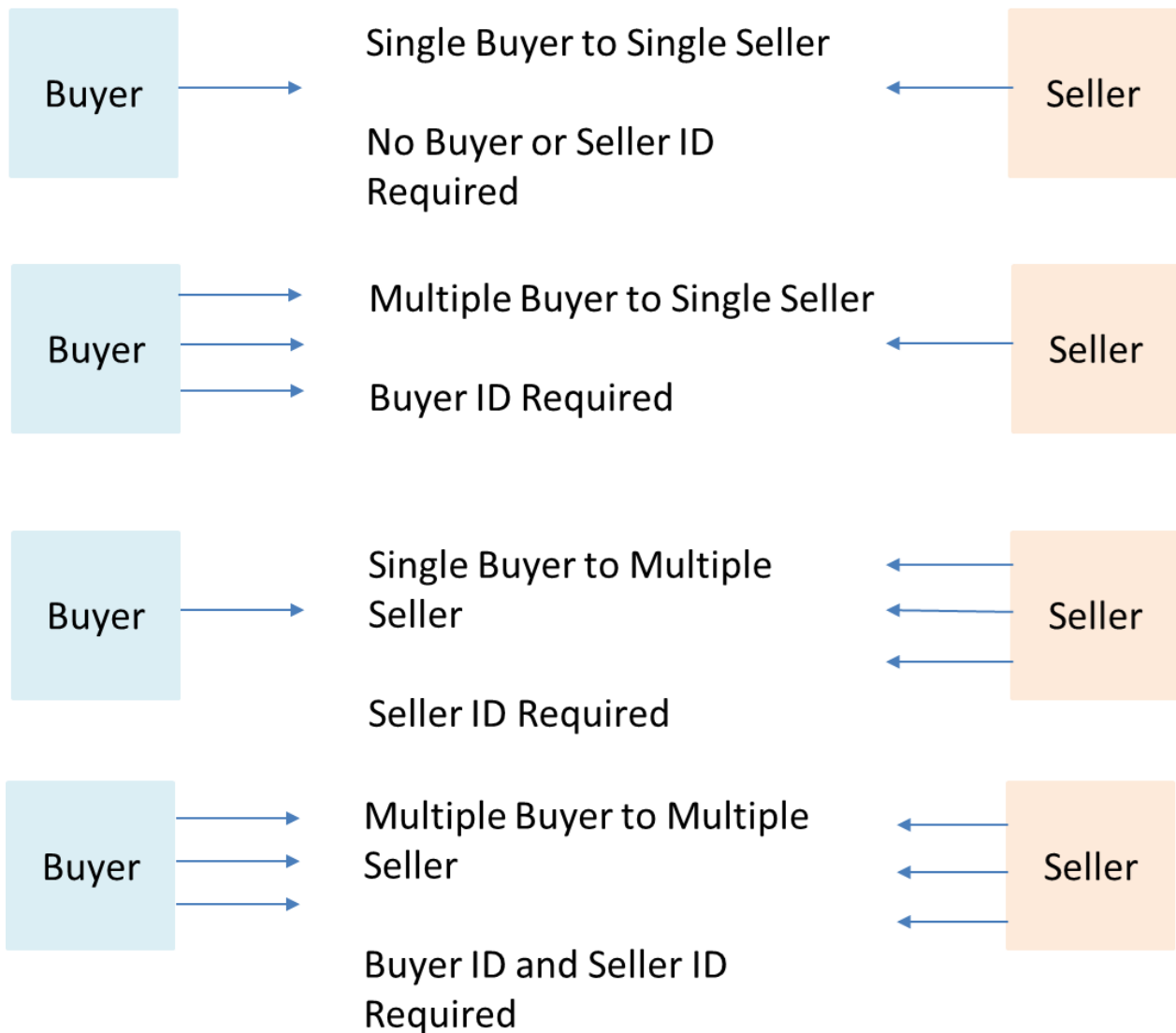


Figure 5. Buyer ID and Seller ID Examples

As shown in Figure 5, if a Requesting Entity representing a single Buyer is doing business with a Responding Entity representing a single Seller, Buyer and Seller IDs are not required to be passed between the two Entities. If a Requesting Entity representing more than one Buyer is doing business with a Responding Entity representing a single Seller, the Buyer ID is required to be passed between the two Entities. If a Requesting Entity representing a single Buyer is doing business with a Responding Entity representing multiple Sellers, the Seller ID is required to be passed between the two Entities. If a Requesting Entity representing multiple Buyers is doing business with a Responding Entity representing multiple Sellers, both the Buyer ID and the Seller ID are required to be passed between the Entities.

While it is outside the scope of this specification, it is assumed that the Requesting Entity and the Responding Entity are aware of each other and can authenticate requests initiated by the other party. It is further assumed that the Buying Entity knows:

- the list of Buyers the Requesting Entity represents when interacting with this Responding Entity; and
- the list of Sellers that this Responding Entity represents to this Requesting Entity.

It is also assumed that the Responding Entity knows:

- the list of Sellers that this Responding Entity represents to this Requesting Entity and

- the list of Buyers the Requesting Entity represents when interacting with this Responding Entity.

In the API the **buyerId** and **sellerId** are represented as optional query parameters in each operation defined.

[R2] If the Requesting Entity has the authority to represent more than one Buyer the request **MUST** include **buyerId** that identifies the Buyer being represented [MEF150 R62]

[R3] If the Responding Entity represents more than one Seller to this Buyer the request **MUST** include **sellerId** that identifies the Seller with whom this request is associated [MEF150 R63]

5.3. API Resource Schema Summary

This subchapter describes the resource model used by the API.

Each entity is a simple or composed type (with the use of **allOf** keyword for data types composition). A simple type defines a set of properties that might be of an object, primitive, or reference type.

[Section 6](#) provides examples of data model and API usage. For a detailed description of the data model, please refer to [API Details](#).

Figure 6 presents the data model for use cases 4 and 5 (not including Errors).

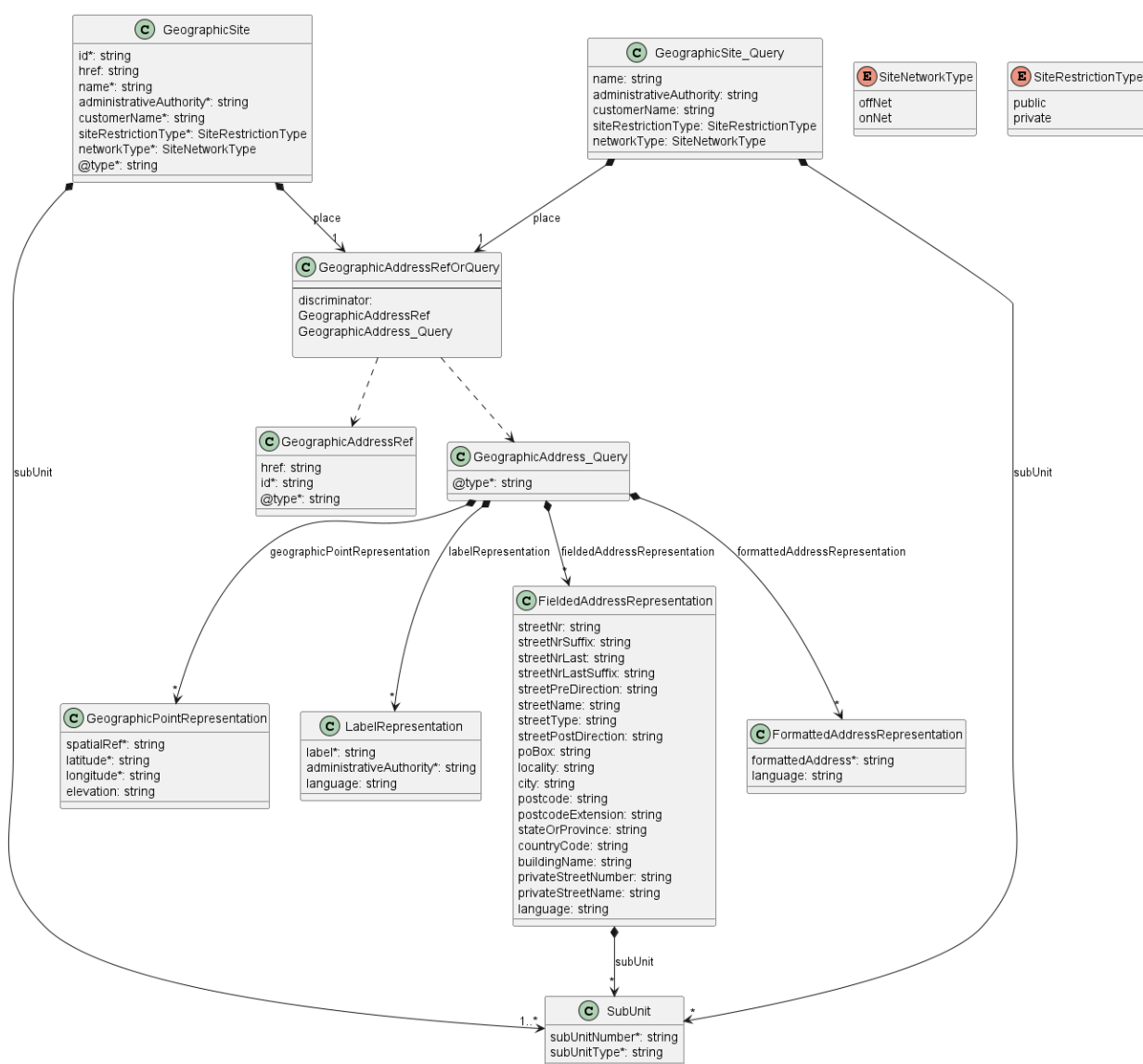


Figure 6. Data Model

While showing the extends relation, for clarity, the extending type lists only its own attributes without the inherited ones.

The root entity is the **GeographicSite**. It is used by both endpoints. Apart from a few simple attributes, it defines referential attributes:

- **place** - a set of location descriptions, each of which describes where this Geographic Site is located.
- **subUnit** - to describe the Site's detailed position within the Geographic Address.

The **place** can be specified with the use of **GeographicAddressRefOrQuery**. This means that the **GeographicAddress** can either be referred to by its **id**, or by value of at least one of the Geographic Address representations.

A **Site** cannot be a part of another **Site** nor describe its **place**. That is why **GeographicAddressRefOrQuery** is used instead of standard **PlaceRefOrValue** - to ensure that a **Site** will not be used as a referenced type.

There are four types of Geographic Address Representations:

- **FieldedAddressRepresentation**
- **FormattedAddressRepresentation**
- **LabelRepresentation**
- **GeographicPointRepresentation**

The **GeographicAddress** model together with its above-mentioned representations and respective requirements are defined by [MEF 121.1](#) (chapter 5.3). That standard is the owner of those definitions. This API specification contains a model of **GeographicAddress** but does not define it. Any further changes of these types will update the API specification, but will not be reflected in this document.

The mandatory **@type** attribute of **GeographicAddressRef** and **GeographicAddress_Query** is used as a discriminator to unambiguously identify the intended type when using in the context of the **oneOf** section of **GeographicAddressRefOrQuery** type.

5.4. Model Structural Validation

The structure of the HTTP payloads exchanged via Site Retrieval API endpoints is defined in the OpenAPI document which is an integral part of this standard.

[R4] Implementations **MUST** use payloads that conform to these definitions.

[R5] The Buyer and the Seller **MUST NOT** use any operation, entity or attribute that is not explicitly defined or allowed by this standard.

5.5. Security Considerations

There must be an authentication mechanism whereby a Seller can be assured who a Buyer is and vice-versa. There must also be authorization mechanisms in place to control what a particular Buyer or Seller is allowed to do and what information may be obtained. However, the definition of the exact security mechanism and configuration is outside the scope of this document. Security considerations are standardized by *LSO API Security Profile* [[MEF 128.1](#)].

6. API Interaction & Flows

This section provides a detailed insight into the API functionality, use cases, and flows. First, a list of business use cases is presented, followed by examples and an explanation of all usage aspects involved.

Table 4 keeps the original use case numbering and naming from MEF 150. The descriptions use API naming.

Use Case #	Use Case Name	Use Case Description
4	Retrieve Service Site List	The Buyer requests that the Seller provides a list of GeographicSites known to the Seller based on filter criteria. For each GeographicSite returned, the Seller also provides a GeographicSite.id , which uniquely identifies it within the Seller.
5	Retrieve Service Site by Service Site Identifier	The Buyer requests the full details for a single GeographicSite based on a GeographicSite.id .

Table 4. Use cases description

The term "Seller Response Code" used in the Business Requirements maps to HTTP response code, where **2xx** indicates *Success* and **4xx** or **5xx** indicate *Failure*.

6.1. Use case 4: Retrieve Service Site List

The flow is a simple request-response pattern, as presented in Figure 7:

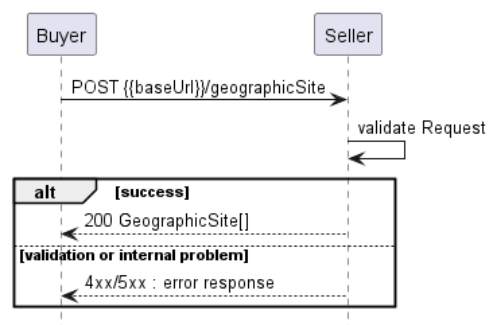


Figure 7. Use case 4: Retrieve Service Site List flow

The Buyer can retrieve a list of **GeographicSites** by using a **POST /geographicSite** operation with desired filtering criteria. The POST operation is used because of the complex query parameters structure that is carried within the body of the request.

[O1] The Buyer **MAY** use any of the following attributes: [MEF150 O7], [MEF150 O9]

- **name**
- **administrativeAuthority**
- **customerName**
- **siteRestrictionType**
- **networkType**

- subUnit

[O2] The Buyer **MAY** use a combination of attributes to avoid getting an **Error422** with **tooManyRecords** code. [MEF150 O8], [MEF150 O10]

[R6] The Buyer **MUST** specify the **place** as search criteria. [MEF150 R43], [MEF150 R44]

The **place** attribute in the request is of **GeographicAddressRefOrQuery** type. This means that the Buyer may point to **GeographicAddress** by its **id**, using the **GeographicAddressRef** type or with the use of representation values, using **GeographicAddress_Query** type.

[R7] The Buyer and the Seller **MUST** agree whether **GeographicAddressRef** and/or **GeographicAddress_Query** can be used in the query.

[R8] If **GeographicAddress_Query** is agreed, the Buyer and the Seller **MUST** agree what Geographic Address Representations may be included in the request. [MEF150 R1]

The following example shows the usage of **GeographicAddressRef** in the request (Use Case 4a). This queries for Sites located in **GeographicAddress** with **id=00000000-0000-0030-0305-873500002014**, **siteRestrictionType=public**, and **networkType: onNet**.

```
{
  "place": {
    "@type": "GeographicAddressRef",
    "id": "00000000-0000-0030-0305-873500002014"
  },
  "siteRestrictionType": "public",
  "networkType": "onNet"
}
```

After receiving the request, the Seller validates it. Then the Seller attempts to match the Buyer's provided criteria with their own Site information. The determination of what is considered a match for a free text field is at the Seller's discretion. If the request filter criteria match one or more Sites known to the Seller, a response is returned with the detailed **GeographicSite** information.

The Seller's response to such a query may look like below:

```
[
  {
    "@type": "GeographicSite",
    "id": "22223333-0000-0030-0305-019283747566",
    "href": "{{baseUri}}/geographicSite/22223333-0000-0030-0305-019283747566",
    "place": {
      "@type": "GeographicAddressRef",
      "id": "00000000-0000-0030-0305-873500002014"
    },
    "name": "Cracow Wasil Room 1",
    "administrativeAuthority": "Authority Example 1",
    "customerName": "Customer Example 1",
    "siteRestrictionType": "public",
    "networkType": "onNet",
    "subUnit": [
      {
        "subUnitNumber": "room",
        "subUnitType": "1"
      }
    ]
  },
  {
    "@type": "GeographicSite",
    "id": "22223333-0000-0030-0305-667584930201",
    "href": "{{baseUri}}/geographicSite/22223333-0000-0030-0305-667584930201",
    "place": {
```

```

    "@type": "GeographicAddressRef",
    "id": "00000000-0000-0030-0305-873500002014"
  },
  "name": "Cracow Wasil Room 2",
  "administrativeAuthority": "Authority Example 1",
  "customerName": "Customer Example 2",
  "siteRestrictionType": "public",
  "networkType": "onNet",
  "subUnit": [
    {
      "subUnitNumber": "room",
      "subUnitType": "2"
    }
  ]
}
]

```

The Seller has found 2 Sites matching the criteria. They are located in **room 1**, and **room 2** as specified by the **subUnit** respectively.

An example below shows an equivalent request with the use of **GeographicAddress_Query** (Use Case 4b). The assumption is that it resolves to exactly the same **GeographicAddress** as in the example above.

```

{
  "place": {
    "@type": "GeographicAddress_Query",
    "fieldedAddressRepresentation": [
      {
        "streetType": "st.",
        "streetName": "Edmunda Wasilewskiego",
        "streetNr": "20",
        "city": "Cracow",
        "stateOrProvince": "Lesser Poland",
        "postcode": "30-305",
        "countryCode": "pl",
        "subUnit": [
          {
            "subUnitType": "floor",
            "subUnitNumber": "4"
          },
          {
            "subUnitType": "apartment",
            "subUnitNumber": "14"
          }
        ],
        "language": "en"
      }
    ]
  },
  "siteRestrictionType": "public",
  "networkType": "onNet"
}

```

This time the Buyer uses **GeographicAddress_Query** and chooses to provide **fieldedAddressRepresentation**. This case should only be used in cases when the Seller does not support identifiers for **GeographicAddresses**.

A Seller's response may look like the following:

```

[
  {
    "@type": "GeographicSite",
    "id": "22223333-0000-0030-0305-019283747566",
    "href": "{{baseUri}}/geographicSite/22223333-0000-0030-0305-019283747566",
    "place": {
      "@type": "GeographicAddress_Query",
      "fieldedAddressRepresentation": [

```

```

    {
      "streetType": "st.",
      "streetName": "Edmunda Wasilewskiego",
      "streetNr": "20",
      "city": "Cracow",
      "stateOrProvince": "Lesser Poland",
      "postcode": "30-305",
      "countryCode": "pl",
      "subUnit": [
        {
          "subUnitType": "floor",
          "subUnitNumber": "4"
        },
        {
          "subUnitType": "apartment",
          "subUnitNumber": "14"
        }
      ],
      "language": "en"
    }
  ],
  "geographicPointRepresentation": [
    {
      "spatialRef": "EPSG:4326",
      "latitude": "50.048868",
      "longitude": "19.929523"
    }
  ]
},
"name": "Cracow Wasil Room 1",
"administrativeAuthority": "Authority Example 1",
"customerName": "Customer Example 1",
"siteRestrictionType": "public",
"networkType": "onNet",
"subUnit": [
  {
    "subUnitNumber": "room",
    "subUnitType": "1"
  }
]
},
{
  "@type": "GeographicSite",
  "id": "22223333-0000-0030-0305-667584930201",
  "href": "{{baseUri}}/geographicSite/22223333-0000-0030-0305-667584930201",
  "place": {
    //same as in the first Site
  },
  "name": "Cracow Wasil Room 2",
  "administrativeAuthority": "Authority Example 1",
  "customerName": "Customer Example 2",
  "siteRestrictionType": "public",
  "networkType": "onNet",
  "subUnit": [
    {
      "subUnitNumber": "room",
      "subUnitType": "2"
    }
  ]
}
]
}
]

```

The Seller responds with the same Sites. Just the **place** is referred to in different ways. Notice that in the response also the **geographicPointRepresentation** is present. The Seller may add this information. The **subUnit** attribute of the request refers only to the **subUnit** attribute of the **GeographicSite** object, not to the **subUnit** of the **GeographicAddress**. In the example above the address specifies **floor 4** and **apartment 14** and that is the precision of that address. The additional placement information is added by the **GeographicSites** at the room level. There are 2 Sites in two separate rooms (**1** and **2**). Thus a valid **subUnit** value in the query would be **room 1** that will filter only the Site with **name="Cracow Wasil Room 1"**. A query with **subUnit** of **floor 4** will result in an empty list.

All requirements apply to both Use Cases 4a and 4b:

[R9] A **GeographicSite** included in the Seller's response **MUST** be located at a **GeographicAddress** having an exact match to the value provided in the **place** attribute of the request. [MEF150 R45]

[R10] If the Buyer provided Geographic Address representations do not match a single **GeographicAddress** in the Seller's system, the Seller **MUST** return an empty list. [MEF150 R46]

[R11] In case no matching records are found, the Seller **MUST** return a valid **200** response with an empty list.

Not having any **GeographicSite** provided for the given Address does not indicate if the Seller is able to serve any type of Product there. For the Buyer to proceed with later steps, it is sufficient to use the **GeographicAddress**.

[R12] In case of too many matching records are found (the definition of 'too many' is up to Seller's discretion), the Seller **MUST** return an **Error422** with **code** equal to **tooManyRecords**.

The Buyer may also ask for pagination of the response when the number of results is too big. The following query attributes related to pagination can be provided:

- **limit** - number of expected list items
- **offset** - offset of the first element in the result list

```
https://serverRoot/mefApi/sonata/geographicSiteManagement/v8/geographicSite?limit=10&offset=0
```

The example above shows a Buyer's request to get the first ten Geographic Sites from a possible list within the response.

The Seller returns a list of elements that comply with the requested **limit**. If the requested **limit** is higher than the supported list size then the smaller list of results is returned. In that case, the size of the result is returned in the header attribute **X-Result-Count**. The Seller can indicate that there are additional results available using:

- **X-Total-Count** header attribute with the total number of available results
- **X-Pagination-Throttled** header set to **true**

[R13] For each **GeographicSite** returned, the Seller **MUST** specify following attributes: [MEF150 R48]

- **id**
- **name**
- **administrativeAuthority**
- **customerName**
- **siteRestrictionType**
- **networkType**
- **subUnit**
- **place**

[R14] Each **GeographicSite** **MUST** be assigned an **id** which is unique within the Seller's network. [MEF150 R50], [MEF150 R52]

[R15] If the request used **GeographicAddressRef** in the **place** attribute, the Seller **MUST** use **GeographicAddressRef** in the **place** attribute in the response as well. [MEF150 R21]

[R16] If the request used `GeographicAddress_Query` in the `place`, the Seller **MUST** use `GeographicAddress_Query` in the `place` in the response as well. [MEF150 R21]

6.2. Use case 5: Retrieve Service Site by Identifier

To get detailed up to date information about the Site, the Buyer sends a Retrieve Site by Identifier Request using a `GET /geographicSite/{id}` operation.

Support of this Use Case is optional.

The flow is a simple request-response pattern, as presented in Figure 7:

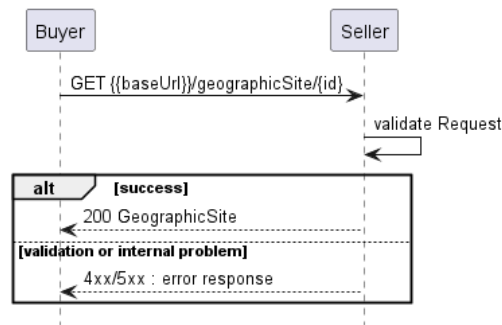


Figure 8. Use case 5: Retrieve Service Site by Identifier flow

Example request and response:

`GET /mefApi/sonata/geographicSiteManagement/v8/geographicSite/22223333-0000-0030-0305-019283747566`

```
{
  "@type": "GeographicSite",
  "id": "22223333-0000-0030-0305-019283747566",
  "href": "{{baseUrl}}/geographicSite/22223333-0000-0030-0305-019283747566",
  "place": {
    "@type": "GeographicAddressRef",
    "id": "00000000-0000-0030-0305-873500002014"
  },
  "name": "Cracow Wasil Room 1",
  "administrativeAuthority": "Authority Example 1",
  "customerName": "Customer Example 1",
  "siteRestrictionType": "public",
  "networkType": "onNet",
  "subUnit": [
    {
      "subUnitNumber": "room",
      "subUnitType": "1"
    }
  ]
}
```

[R17] When providing the response, the Seller **MUST** specify: [MEF150 R54]

- `id`
- `name`
- `administrativeAuthority`
- `customerName`
- `siteRestrictionType`
- `networkType`
- `subUnit`
- `place`

[R18] The Buyer and Seller **MUST** agree on whether the **GeographicAddressRef** or **GeographicAddress_Query** will be used to describe the **place**. [MEF150 R55]

[R19] In case **id** does not find a matching **GeographicSite** in Seller's system, an error response **404** **MUST** be returned.

7. API Details

7.1. API patterns

7.1.1. Indicating errors

Erroneous situations are indicated by appropriate HTTP responses. An error response is indicated by HTTP status 4xx (for client errors) or 5xx (for server errors) and appropriate response payload. The Site retrieval API uses the error responses depicted and described below.

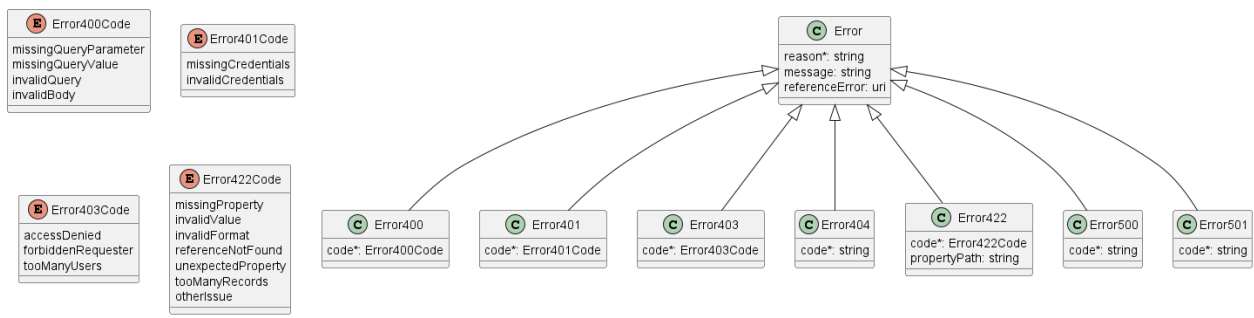


Figure 9. Data model types to represent an erroneous response

7.1.1.1. Type Error

Description: Standard Class used to describe API response error Not intended to be used directly. The `code` in the HTTP header is used as a discriminator for the type of error returned in runtime.

Name	Type	Description
reason*	string maxLength = 255	Text that explains the reason for the error. This can be shown to a client user.
message	string	Text that provides mode details and corrective actions related to the error. This can be shown to a client user.
referenceError	uri format = uri	url pointing to documentation describing the error

7.1.1.2. Type Error400

Description: Bad Request. (<https://tools.ietf.org/html/rfc7231#section-6.5.1>)

Inherits from:

- [Error](#)

Name	Type	Description
code*	Error400Code	

7.1.1.3. enum Error400Code

Description: One of the following error codes:

- `missingQueryParameter`: The URI is missing a required query-string parameter
- `missingQueryValue`: The URI is missing a required query-string parameter value
- `invalidQuery`: The query section of the URI is invalid.
- `invalidBody`: The request has an invalid body

7.1.1.4. Type Error401

Description: Unauthorized. (<https://tools.ietf.org/html/rfc7235#section-3.1>)

Inherits from:

- [Error](#)

Name	Type	Description
code*	Error401Code	

7.1.1.5. `enum` Error401Code

Description: One of the following error codes:

- `missingCredentials`: No credentials provided.
- `invalidCredentials`: Provided credentials are invalid or expired

7.1.1.6. Type Error403

Description: Forbidden. (<https://tools.ietf.org/html/rfc7231#section-6.5.3>)

Inherits from:

- [Error](#)

Name	Type	Description
code*	Error403Code	

7.1.1.7. `enum` Error403Code

Description: This code indicates that the server understood the request but refuses to authorize it because of one of the following error codes:

- `accessDenied`: Access denied
- `forbiddenRequester`: Forbidden requester
- `tooManyUsers`: Too many users

7.1.1.8. Type Error404

Description: Resource for the requested path not found. (<https://tools.ietf.org/html/rfc7231#section-6.5.4>)

Inherits from:

- [Error](#)

Name	Type	Description
------	------	-------------

code*	string	The following error code: - notFound: A current representation for the target resource not found
-------	--------	--

7.1.1.9. Type Error422

The response for HTTP status **422** is a list of elements that are structured using the **Error422** data type. Each list item describes a business validation problem. This type introduces the **propertyPath** attribute which points to the erroneous property of the request, so that the Buyer may fix it easier. It is highly recommended that this property should be used, yet remains optional because it might be hard to implement.

Description: Unprocessable entity due to a business validation problem. (<https://tools.ietf.org/html/rfc4918#section-11.2>)

Inherits from:

- [Error](#)

Name	Type	Description
------	------	-------------

code*	Error422Code	
-------	------------------------------	--

propertyPath	string	A pointer to a particular property of the payload that caused the validation issue. It is highly recommended that this property should be used. Defined using JavaScript Object Notation (JSON) Pointer (https://tools.ietf.org/html/rfc6901).
--------------	--------	--

7.1.1.10. enum Error422Code

Description: One of the following error codes:

- missingProperty: The property the Seller has expected is not present in the payload
- invalidValue: The property has an incorrect value
- invalidFormat: The property value does not comply with the expected value format
- referenceNotFound: The object referenced by the property cannot be identified in the Seller system
- unexpectedProperty: Additional property, not expected by the Seller has been provided
- tooManyRecords: the number of records to be provided in the response exceeds the Seller's threshold.
- otherIssue: Other problem was identified (detailed information provided in a reason)

7.1.1.11. Type Error500

Description: Internal Server Error. (<https://tools.ietf.org/html/rfc7231#section-6.6.1>)

Inherits from:

- [Error](#)

Name	Type	Description
------	------	-------------

Name	Type	Description
------	------	-------------

code*	string	The following error code: - <code>internalError</code> : Internal server error - the server encountered an unexpected condition that prevented it from fulfilling the request.
-------	--------	--

7.1.1.12. Type Error501

Description: Not Implemented. (<https://tools.ietf.org/html/rfc7231#section-6.6.2>)

Inherits from:

- [Error](#)

Name	Type	Description
------	------	-------------

code*	string	The following error code:: - <code>notImplemented</code> : Method not supported by the server
-------	--------	---

7.2. API Data model

Figure 9 presents the Site Retrieval data model. The data types, requirements related to them, and mapping to MEF 79 and MEF 79.0.2 specifications are discussed later in this section.

This data model is used to construct requests and responses of the API endpoints described in [Section 5.2.1](#)

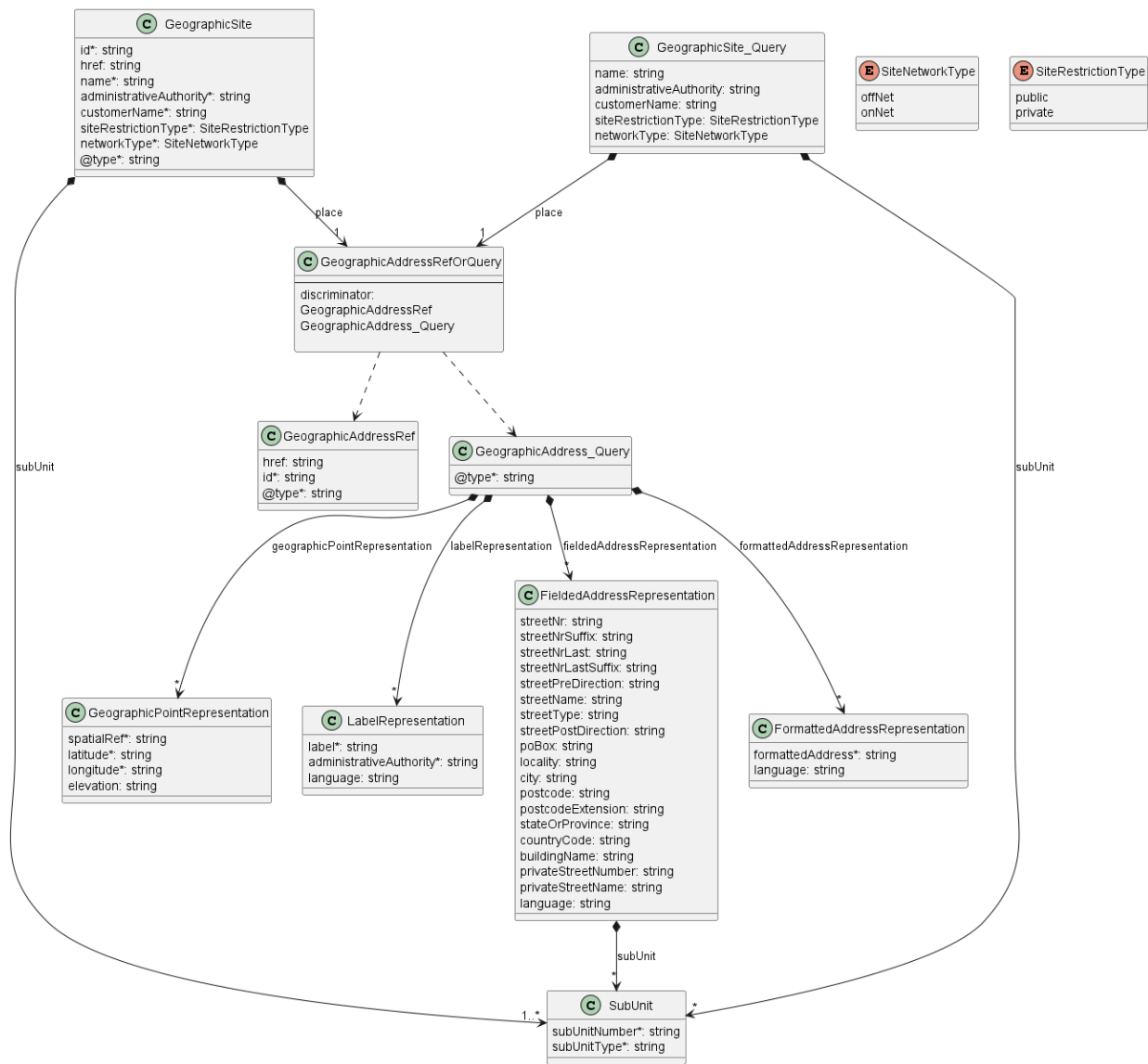


Figure 10. Site Retrieval Data Model

7.2.1. Geographic Site

7.2.1.1 Type GeographicSite

Description: A fixed or mobile place at which a Product can be installed. A Site is always associated with a Geographic Address. This association may include Sub-Units describing where precisely within that Geographic Address this particular Geographic Site is located.

Name	Type	M/O	Description	MEF 150
id	string	M	Identifier of the Geographic Site unique within the Seller.	Service Site Identifier
href	string	O	Unique reference of the Geographic Site unique within the Seller.	Not represented in MEF 79

Name	Type	M/O	Description	MEF 150
place	GeographicAddressRefOrQuery	M	A Geographic Address described by identifier or by a set of Geographic Address Representations.	Service Site Location
name	string	M	The name for the Service Site.	Service Site Name
administrativeAuthority	string	M	The name of the company that is the administrative authority (e.g., controls access) for this Service Site. (For example, the building owner)	
customerName	string	M	The name of the company that is the administrative authority for the space within this Service Site. (For example, the company leasing space in a multi-tenant building).	Service Site Customer Name
siteRestrictionType	SiteRestrictionType	M	This defines whether a Geographic Site is public or private. public - means that the existence of this Geographic Site is public information. A meet-me-room in a hosted data center facility (where all interconnects between parties take place) is an example of a	

Name	Type	M/O	Description	MEF 150
			public Geographic Site. A shared facility in the basement of a multi-tenant business building where all interconnects between parties take place is another example of a public Geographic Site. private - means that the existence of this Geographic Site is on a need-to-know basis. A wiring closet set up inside a customer facility just to connect two parties is an example of a private Geographic Site. For private sites, the Seller does not return any information regarding the existence of this Geographic Site unless it has been established that this Buyer is authorized to obtain this information.	

Name	Type	M/O	Description	MEF 150
networkType	SiteNetworkType	M	Indicates whether the Seller is owner of the network to which the Service Site is connected or whether that network is owned by a 3rd party.	
subUnit	SubUnit[]	M	A list of Sub Units which describes the Service Site detailed position within the Geographic Address. This is a list to allow complex sub-unit information such as SUITE 42 ROOM A	
@type	string	M	Used to unambiguously designate the class type when using `oneOf`	Not represented in MEF 79

7.2.1.2 Type GeographicSite_Query

Description: A flavor of the [GeographicSite](#) type to be used in the context of the [listGeographicSite](#) operation. It does not contain the [id](#), [href](#), and [@type](#) attributes and all the remaining, despite [place](#), are optional.

Name	Type	M/O	Description	MEF 150
------	------	-----	-------------	---------

Name	Type	M/O	Description	MEF 150
place	GeographicAddressRefOrQuery	M	A set of location descriptions, each of which describes where this GeographicSite is located. It is important to note that this is a set because a particular Geographic Site might be described with multiple locations. For example, one Geographic Site might have two Fielded Addresses (for a building on the corner of two streets), two Formatted Addresses, and a Geographic Point.	
name	string	O	The name for the Service Site.	
administrativeAuthority	string	O	The name of the company that is the administrative authority (e.g., controls access) for this Service Site. (For example, the building owner)	
customerName	string	O	The name of the company that is the administrative authority for the space within this Service Site. (For example, the company leasing space in a multi-tenant building).	

Name	Type	M/O	Description	MEF 150
siteRestrictionType	SiteRestrictionType	O	This defines whether a Geographic Site is public or private. public - means that the existence of this Geographic Site is public information. A meet-me-room in a hosted data center facility (where all interconnects between parties take place) is an example of a public Geographic Site. A shared facility in the basement of a multi-tenant business building where all interconnects between parties take place is another example of a public Geographic Site. private - means that the existence of this Geographic Site is on a need-to-know basis. A wiring closet set up inside a customer facility just to connect two parties is an example of a private Geographic Site. For private sites, the Seller does not return any information regarding the existence of this Geographic Site unless it has been established that this Buyer is authorized to obtain this information.	

Name	Type	M/O	Description	MEF 150
networkType	SiteNetworkType	O	Indicates whether the Seller is owner of the network to which the Service Site is connected or whether that network is owned by a 3rd party.	
subUnit	SubUnit[]	O	A list of Sub Units which describes the Service Site detailed position within the Geographic Address. This is a list to allow complex sub-unit information such as SUITE 42 ROOM A	

7.2.1.3. **enum** SiteRestrictionType

Description: This defines whether a Geographic Site is public or private.

public - means that the existence of this Geographic Site is public information. A meet-me-room in a hosted data center facility (where all interconnects between parties take place) is an example of a public Geographic Site. A shared facility in the basement of a multi-tenant business building where all interconnects between parties take place is another example of a public Geographic Site.

private - means that the existence of this Geographic Site is on a need-to-know basis. A wiring closet set up inside a customer facility just to connect two parties is an example of a private Geographic Site. For private sites, the Seller does not return any information regarding the existence of this Geographic Site unless it has been established that this Buyer is authorized to obtain this information.

Value **MEF 150**

public PUBLIC

private PRIVATE

7.2.1.4. **enum** SiteNetworkType

Description: Indicates whether the Seller is owner of the network to which the Service Site is connected or whether that network is owned by a 3rd party.

Value **MEF 150**

offNet OFF_NET

Value **MEF 150**

onNet ON_NET

7.2.2. Geographic Address

7.2.2.1 Type GeographicAddressRefOrQuery

Description: Allows pointing to a [GeographicAddress](#) either by reference or by value.

7.2.2.2 Type GeographicAddressRef

Description: A reference to a Geographic Address resource available through Address Validation API.

Name	Type	M/O	Description	MEF 150
href	string	O	Hyperlink to the referenced Address. Hyperlink MAY be used by the Seller in responses. Hyperlink MUST be ignored by the Seller in case it is provided by the Buyer in a request.	Not represented in MEF 79
id	string	M	Identifier of the referenced Geographic Address. This identifier is assigned during a successful address validation request (Geographic Address Management API)	Fielded Formatted Geographic Address Label Geographic Point Identifier
@type	string	M	Used to unambiguously designate the class type when using `oneOf`	

7.2.2.3. Type GeographicAddress_Query

Description: A list of representations being a subset of Geographic Address entity. This is to be used when providing a list of representations to validate or search for an Geographic Address

Name	Type	M/O	Description	MEF 150
fieldedAddressRepresentation	FieldedAddressRepresentation[]	O	A Fielded Address representation	
formattedAddressRepresentation	FormattedAddressRepresentation[]	O	A Formatted Address representation	
geographicPointRepresentation	GeographicPointRepresentation[]	O	A Fielded Address representation	
labelRepresentation	LabelRepresentation[]	O	A Fielded Address representation	

Name	Type	M/O	Description	MEF 150
@type	string	M	Used to unambiguously designate the class type when using `oneOf`	

7.2.2.4. Type FieldedAddressRepresentation

Description: A type of Address that has a discrete field and value for each type of boundary or identifier down to the lowest level of detail. For example "street number" is one field, "street name" is another field, etc.

Name	Type	M/O	Description	MEF 150
streetNr	string	O	Number identifying a specific property on a public street. It may be combined with streetNrLast for ranged addresses.	
streetNrSuffix	string	O	The first street number suffix (in a street number range) or the suffix for the street number if there is no range	
streetNrLast	string	O	Last number in a range of street numbers allocated to an Address	
streetNrLastSuffix	string	O	Last street number suffix for a ranged Address	
streetPreDirection	string	O	The direction of the street that appears before the Street Name	
streetName	string	O	Name of the street or other street type	
streetType	string	O	The type of street (e.g., alley, avenue, boulevard, brae, crescent, drive, highway, lane, terrace, parade, place, tarn, way, wharf)	
streetPostDirection	string	O	A modifier denoting a relative direction that appears after the Street Name.	
poBox	string	O	Number identifying a specific location in a post office.	
locality	string	O	An area of defined or undefined boundaries within a local authority or other legislatively defined area.	
city	string	O	City in which the Address is located.	
postcode	string	O	A descriptor for a postal delivery area used to speed and simplify the delivery of mail (also known as zip code)	
postcodeExtension	string	O	The extension used on a postal code. Note: there are different use codes for this attribute depending upon the country.	

Name	Type	M/O	Description	MEF 150
stateOrProvince	string	O	The State or Province in which the Address is located.	
countryCode	string minLength = 2 maxLength = 2	O	Country in which the Address is located, defined using two characters as defined in ISO 3166	
subUnit	SubUnit[]	O	The Service Site Sub Unit represented as a list. This is a list to allow complex sub-unit information such as SUITE 42 ROOM A	
buildingName	string	O	The well-known name of a building that is located at this Address (e.g., where there is one Address for a campus).	
privateStreetNumber	string	O	Street number on a private street within the Address.	
privateStreetName	string	O	Private streets internal to a property (e.g., a university) may have internal names that are not recorded by the land title office.	
language	string minLength = 2 maxLength = 2	O	The language in which the address is expressed. Based on ISO 639:2023	

7.2.2.5. Type FormattedAddressRepresentation

Description: A freeform text representation agreed to by the Buyer and Seller.

Name	Type	M/O	Description	MEF 150
formattedAddress	string	M	A formatted Address Representation that contains a non-fielded address.	
language	string minLength = 2 maxLength = 2	O	The language in which the address is expressed. Based on ISO 639:2023	

7.2.2.6. Type GeographicPointRepresentation

Description: A GeographicPointRepresentation defines a geographic point through coordinates.

Name	Type	M/O	Description	MEF 150
spatialRef	string	M	The spatial reference system used to determine the coordinates. The system used and the value of this field are to be agreed during the onboarding process.	
latitude	string	M	The latitude expressed in the format specified by the 'spacialRef'	

Name	Type	M/O	Description	MEF 150
longitude	string	M	The longitude expressed in the format specified by the `spacialRef`	
elevation	string	O	The elevation expressed in the format specified by the `spacialRef`	

7.2.2.7. Type LabelRepresentation

Description: A unique identifier controlled by a generally accepted independent administrative authority that specifies a fixed geographical location.

Name	Type	M/O	Description	MEF 150
label	string	M	The unique reference to an Geographic Address assigned by the Administrative Authority.	
administrativeAuthority	string	M	The organization or standard from the organization that administers this Geographic Address Label ensuring it is unique within the Administrative Authority.	
language	string minLength = 2 maxLength = 2	O	The language in which the label is expressed. Based on ISO 639:2023	

7.3. Common

7.3.1. Type SubUnit

Description: Allows for sub unit identification

Name	Type	M/O	Description	MEF 150
subUnitNumber	string	M	The discriminator used for the subunit, often just a simple number but may also be a range.	
subUnitType	string	M	The type of subunit e.g. BERTH, FLAT, PIER, SUITE, SHOP, TOWER, UNIT, WHARF.	

8. References

- [MEF 50.1](#), MEF Services Lifecycle Process Flows, August 2017.
- [MEF 55.1](#), Lifecycle Service Orchestration (LSO): Reference Architecture and Framework, February 2021
- [MEF 55.1.1](#), Amendment to MEF 55.1: Reference Architecture and Framework - Terminology, June 2023
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